Improving Smoothed Wasserstein Error Bounds in Langevin Sampling via Effective Dimension

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***Abstract*—Sampling from high-dimensional distributions is a central problem in Bayesian inference and machine learning, but practical samplers produce dependent samples and may introduce discretization bias. We study the unadjusted Langevin algorithm (ULA) for strongly log-concave targets and analyze the accuracy of the post-burn-in empirical measure using a Gaussian-smoothed Wasserstein-1 distance, which regularizes the test-function class and makes the error analysis more tractable in high dimensions.Our main result is a clean decomposition of the total error into three interpretable parts: burn-in error (distance to the ULA stationary law), discretization bias (difference between the ULA stationary law and the true target), and sampling/dependence error along the Markov chain trajectory. We bound the burn-in term using Wasserstein contraction of ULA and control the sampling/dependence term using Markov-chain concentration together with covering-number arguments for the smoothed test class.To refine the dimension dependence of the sampling term, we introduce a spectral complexity measure based on the covariance structure of the ULA stationary distribution. In the Gaussian case, this yields an effective-dimension improvement over ambient-dimension bounds. For general strongly log-concave targets, we discuss a local quadratic approximation approach and identify the main high-dimensional obstruction to extending the Gaussian refinement beyond near-quadratic regimes.**

I. Introduction

Sampling from high-dimensional probability distributions is a foundational computational task that appears across statistics, machine learning, and the physical sciences. In Bayesian inference, the goal is rarely a single point estimate; rather, we seek posterior expectations, uncertainty quantification, and predictive distributions, all of which are expressed as integrals with respect to a posterior measure. In modern machine learning, sampling underlies energy-based models, probabilistic generative modeling, and decision-making under uncertainty. In these settings, the ambient dimension can be large and the target distribution is typically specified only up to a normalization constant. As a result, direct sampling procedures (such as inverse transform sampling or rejection sampling) are either unavailable or become computationally infeasible as dimension grows, motivating the study of scalable, theoretically grounded sampling algorithms.

A particularly important and widely studied class of targets are *log-concave* distributions of the form $\pi(x) \propto e^{-V(x)}$ where the potential V is convex. Log-concavity captures many posterior distributions in generalized linear models (e.g., logistic regression with a Gaussian prior), and it provides a regime in which the geometry of the target is well-behaved: there is no severe multimodality, and functional inequalities can be used to derive quantitative convergence rates. These properties make log-concave sampling a natural starting point for developing non-asymptotic theory that is both mathematically clean and practically relevant.

Among the most influential sampling methods in this regime are Langevin-based algorithms. The continuous-time Langevin diffusion defines a stochastic process whose invariant distribution is precisely the target π, and it enjoys strong convergence guarantees under standard convexity and smoothness assumptions. In computation, however, one must discretize the diffusion, leading to practical algorithms such as the unadjusted Langevin algorithm (ULA), also known as Langevin Monte Carlo (LMC). This discretization is attractive because it resembles noisy gradient descent and requires only evaluations of ∇V, making it scalable to large dimensions and large datasets (with further variants such as stochastic-gradient Langevin dynamics). Yet discretization comes at a cost: unlike the diffusion, the discretized chain typically has its own stationary distribution, which differs from π unless the step size is taken sufficiently small. At the same time, even when targeting the correct stationary law, samples produced by Markov chain methods are dependent, so empirical averages and empirical measures exhibit variance inflation relative to the ideal i.i.d. setting. These issues are not technicalities; they are the practical bottlenecks that determine how much compute is required for reliable inference.

This project focuses on understanding and quantifying the accuracy of Langevin-based sampling through the lens of *empirical (occupation) measures*. Rather than studying only bias in expectations of a fixed test function, we ask a stronger question: how close is the entire empirical distribution of the generated samples to the target distribution? Measuring this discrepancy is subtle in high dimension. Classical distances between measures (including Wasserstein-1) can behave pessimistically for empirical measures, and their rates often degrade sharply with dimension due to the richness of the class of Lipschitz test functions. Motivated by this, we adopt a *smoothed Wasserstein* metric, which compares Gaussian-regularized versions of the measures and therefore controls a smaller, better-behaved function class. This choice provides a principled middle ground: the metric remains sensitive to distributional mismatch while allowing sharper complexity control than unsmoothed Wasserstein distances.

The motivation for selecting this topic is twofold. First, Langevin-type methods are increasingly used as workhorses

for high-dimensional Bayesian computation and as building blocks in modern generative modeling, so understanding their finite-sample behavior is directly relevant to practice. Second, there is an active research frontier in developing *non-asymptotic* guarantees that separate and quantify the distinct sources of error in practical sampling pipelines: burn-in (convergence to stationarity), discretization bias (difference between the diffusion target and the algorithm's invariant law), and finite-sample fluctuations under dependence. A clear decomposition of these effects, coupled with dimension-aware complexity bounds, provides actionable guidance for algorithm design choices such as step size, burn-in length, and total sampling budget.

In this report, we review the Langevin diffusion and its convergence properties for strongly log-concave targets, introduce Gaussian-smoothed Wasserstein distance as a tool for assessing empirical-measure accuracy in high dimension, and develop an error decomposition that isolates burn-in, discretization, and dependent sampling contributions. We further discuss refined dimension dependence in Gaussian settings through an effective-dimension viewpoint and outline the obstacles to extending such refinements to general log-concave targets beyond local quadratic approximations. Overall, the goal is to provide a coherent theoretical picture that links the geometry of V, the dynamics of Langevin sampling, and the statistical behavior of empirical measures.

II. Background and Literature on Langevin Sampling and Wasserstein Errors

We consider a target distribution π on ^{d} given in the energy form

$$\pi(x) = \frac{e^{-V(x)}}{Z}, \qquad Z := \int_d e^{-V(x)}\,dx, \tag{1}$$

where $V\ :^d\to$ is the potential and Z is the normalizing constant. The computational goal of *sampling* is to generate $X \sim \pi$, primarily because sampling enables Monte Carlo evaluation of high-dimensional integrals:

$$_\pi[f(X)] = \int_d f(x)\,\pi(x)\,dx \approx \frac{1}{n}\sum_{i=1}^{n} f(X_i), \qquad X_i \sim \pi.$$

In many statistical and machine-learning problems (e.g., Bayesian inference), the normalizer Z is intractable, which blocks direct methods such as inverse transforms and typically renders rejection sampling inefficient in high dimension unless the proposal is exceptionally well-matched to π.

A central structural regime is *log-concave sampling*. A distribution is log-concave if $\log \pi(x)$ is concave; since $\log \pi(x) = -V(x) - \log Z$, log-concavity is equivalent to convexity of V. The convex regime is emphasized in the modern sampling literature because it rules out severe multimodality, gives controlled geometry, and enables functional inequalities (Poincaré/log-Sobolev) that yield quantitative mixing and concentration bounds; see [1] for a unified treatment. Stronger assumptions, such as m-strong convexity and L-smoothness,

$$mI_d \preceq \nabla^2 V(x) \preceq LI_d, \qquad \forall x \in^d,$$

lead to particularly clean non-asymptotic guarantees for Langevin-type algorithms [2]–[4].

A canonical sampling mechanism for targets of the form (1) is the Langevin diffusion

$$dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dB_t, \tag{2}$$

where B_t is standard Brownian motion in ^{d}. Under standard regularity assumptions, π is invariant for (2), and under strong log-concavity the law of X_t converges to π exponentially fast (e.g., in KL divergence via entropy dissipation and log-Sobolev inequalities); see Appendix **??** and [1] for background.

Despite its conceptual clarity, (2) is a continuous-time object and cannot be simulated exactly on a computer. A concrete obstruction is the roughness of Brownian motion: almost surely, B_t is nowhere differentiable, and the finite-difference slope $(B_{t+h} - B_t)/h$ has variance $1/h$ and blows up as $h \to 0$ (Appendix A-A). This forces time discretization. The standard discretization is the Euler–Maruyama scheme, leading to the unadjusted Langevin algorithm (ULA/LMC):

$$X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0, I_d). \tag{3}$$

Non-asymptotic analyses of ULA in the strongly log-concave regime are developed in [2]–[4].

A key point in the ULA literature is that discretization generally changes the invariant distribution. Let π_h denote the stationary distribution of the Markov chain (3). In general,

$$\pi_h \neq \pi \qquad \text{for any fixed } h > 0,$$

so ULA carries an intrinsic *discretization bias*. This is already visible in the simplest quadratic case $V(x) = \frac{1}{2}\|x\|^2$, where $\pi = \mathcal{N}(0, I_d)$ but π_h is a different Gaussian (Appendix A-C). Consequently, end-to-end guarantees typically decompose error into: (i) *burn-in* (mixing from initialization to π_h), (ii) *discretization bias* (distance between π_h and π), and (iii) *Monte Carlo error* due to finite-length dependent averaging [2]–[4].

To quantify discretization bias, Wasserstein distances are a standard tool, especially W_2 because it interacts well with synchronous couplings and strong convexity. The literature commonly proves (under strong log-concavity and step-size restrictions) bounds of the form

$$W_2(\pi_h, \pi) \le C(\text{problem parameters}) \cdot \sqrt{d}\,h \quad ,$$

and the dependence on d is unavoidable in general. A widely used proof strategy has a local-to-global flavor: first show that one exact diffusion step of length h and one ULA step starting from the same point are close in W_2 under a shared Brownian increment (a local coupling estimate), then amplify this using

global contractivity of the ULA kernel in W_2 induced by strong convexity (Appendix A-E). In the quadratic Gaussian case, the stationary bias can be computed explicitly in W_2, yielding linear behavior in h for small h and a $\sqrt{d}$ scaling (Appendix A-F).

Beyond stationary bias, one must account for mixing and Monte Carlo fluctuations. For strongly log-concave targets, burn-in is often controlled by W_2 contraction: synchronous coupling cancels the injected noise and reveals a deterministic contraction driven by monotonicity of ∇V under strong convexity. This yields geometric decay of $W_2((X_k), \pi_h)$ as k grows (Appendix A-D). Monte Carlo error for Markov chains is governed by dependence through integrated autocorrelation time and effective sample size, and limit theorems (LLN/CLT) for geometrically ergodic chains formalize this behavior; see standard Markov chain texts for general conditions.

A separate and influential line of work studies *distribution-level* error of empirical and occupation measures in Wasserstein distance, rather than only fixed test functions. For i.i.d. samples, the sharp dimension dependence of $W_p(\mu_n, \mu)$ is by now well understood under moment assumptions; see [5], [6]. For dependent samples generated by Markov chains, additional geometric inputs are needed; representative examples include bounds for occupation measures in W_1 [7] and approaches based on curvature/contraction ideas [8].

A modern technique to mitigate the curse of dimensionality in distribution-level metrics is *smoothing*. Instead of comparing μ and ν directly, one compares their Gaussian-regularized versions $\mu * \mathcal{N}(0, \sigma^2 I)$ and $\nu * \mathcal{N}(0, \sigma^2 I)$, effectively restricting attention to features resolvable at scale σ. Goldfeld, Greenewald, Polyanskiy, and Weed develop a general framework for convergence of *smoothed empirical measures* and connect rates to refined complexity notions and effective dimension [9]. This smoothing viewpoint is particularly natural for Wasserstein-1, since W_1 admits a dual formulation as a supremum over 1-Lipschitz test functions, and smoothing reduces the complexity of the resulting function class.

Finally, controlling empirical processes along a Markov chain often relies on concentration inequalities for dependent sequences. Paulin proves Hoeffding-type concentration bounds for empirical averages of bounded functions of a Markov chain in terms of mixing parameters such as the pseudo-spectral gap [10]. A recurring bottleneck in applications is the boundedness requirement: many natural observables in continuous state space are Lipschitz but unbounded. A standard resolution, compatible with strongly log-concave settings, is truncation: clip the observable to apply a bounded-function inequality, then control the clipping error using sub-Gaussian tails for Lipschitz functions under log-Sobolev concentration. We provide a fully self-contained truncation reduction (the part that is typically hand-waved) in Appendix A-I.

III. Method

A. Problem formulation

Let π be a target probability distribution on $\mathbb{R}^d$ with density

$$\pi(x) \propto e^{-V(x)}.$$

Throughout, we assume that V is strongly convex and smooth:

$$\mu I \preceq \nabla^2 V(x) \preceq LI, \qquad \forall x \in \mathbb{R}^d,$$

for some $0 < \mu \leq L < \infty$ (see, e.g., [3], [11], [12]).

We consider the unadjusted Langevin algorithm (ULA) with step size $h > 0$:

$$X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_{k+1}, \qquad \xi_{k+1} \sim \mathcal{N}(0, I_d). \tag{4}$$

Let π_h denote the invariant law of the ULA chain (when it exists). In general, $\pi_h \neq \pi$ because of discretization bias [3], [12].

After a burn-in period T, we form the empirical (occupation) measure

$$\widehat{\mu}_{n,T} := \frac{1}{n} \sum_{k=T+1}^{T+n} \delta_{X_k}. \tag{5}$$

To quantify approximation error, we use the *smoothed Wasserstein-1* distance. Let

$$\nu_\sigma := \mathcal{N}(0, \sigma^2 I_d),$$

and define

$$W_1^{(\sigma)}(\alpha, \beta) := W_1(\alpha * \nu_\sigma, \beta * \nu_\sigma). \tag{6}$$

We also write φ_σ for the density of ν_σ.

A key motivation is the Kantorovich–Rubinstein dual representation:

$$W_1(\alpha, \beta) = \sup_{\|f\|_{\mathrm{Lip}} \leq 1} \Big(\mathbb{E}_\alpha[f] - \mathbb{E}_\beta[f] \Big), \tag{7}$$

so controlling Wasserstein distance is equivalent to controlling worst-case errors over all 1-Lipschitz test functions [5], [13].

B. Error decomposition

We decompose the total error into burn-in, discretization bias, and sampling/dependence terms. By the triangle inequality,

$$W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \pi) \leq W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \pi_h) + W_1^{(\sigma)}(\pi_h, \pi). \tag{8}$$

Applying the triangle inequality again to the first term gives

$$W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \pi_h) \leq W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T)) + W_1^{(\sigma)}(\mathcal{L}(X_T), \pi_h). \tag{9}$$

Since $W_1^{(\sigma)} \leq W_1 \leq W_2$, we have

$$W_1^{(\sigma)}(\pi_h, \pi) \leq W_2(\pi_h, \pi), \qquad W_1^{(\sigma)}(\mathcal{L}(X_T), \pi_h) \leq W_2(\mathcal{L}(X_T), \pi_h). \tag{10}$$

Under strong convexity and smoothness, ULA is contractive in W_2 (for sufficiently small h), yielding the burn-in bound

$$W_2(\mathcal{L}(X_T), \pi_h) \leq (1 - \mu h)^{T/2}\, W_2(\delta_{x_0}, \pi_h), \tag{11}$$

via synchronous coupling [3], [12].

Moreover, standard ULA discretization theory gives

$$W_2(\pi_h, \pi) \lesssim C(\mu, L)\sqrt{d\,h}, \qquad h \lesssim 1/L, \tag{12}$$

see [3], [11], [12].

Combining (8)–(12), we obtain

$$\begin{aligned} W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \pi) \;\lesssim\; & \underbrace{(1-\mu h)^{T/2} W_2(\delta_{x_0}, \pi_h)}_{\text{burn-in}} \\ & + \underbrace{C(\mu, L)\sqrt{d\,h}}_{\text{discretization bias}} + \underbrace{W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T))}_{\text{sampling + dependence}}. \end{aligned} \tag{13}$$

C. Sampling error as a supremum over smoothed test functions

Applying (7) to $\alpha = \widehat{\mu}_{n,T} * \nu_\sigma$ and $\beta = \mathcal{L}(X_T) * \nu_\sigma$, we obtain

$$\begin{aligned} & W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T)) \\ & \quad = \sup_{\|f\|_{\mathrm{Lip}} \le 1} \left| \frac{1}{n} \sum_{k=T+1}^{T+n} (f * \varphi_\sigma)(X_k) - \mathbb{E}\big[(f * \varphi_\sigma)(X_T)\big] \right|. \end{aligned} \tag{14}$$

Define the smoothed test class

$$\mathcal{G}_\sigma := \{ g = f * \varphi_\sigma : \; \|f\|_{\mathrm{Lip}} \le 1 \}. \tag{15}$$

Then (14) becomes

$$W_1^{(\sigma)}(\widehat{\mu}_{n,T}, \mathcal{L}(X_T)) = \sup_{g \in \mathcal{G}_\sigma} \left| \frac{1}{n} \sum_{k=T+1}^{T+n} g(X_k) - \mathbb{E}\, g(X_T) \right|. \tag{16}$$

Thus, the sampling problem reduces to a uniform deviation bound over $\mathcal{G}_\sigma$.

D. Covering argument and finite-class concentration

We control the supremum in (16) via a covering argument.

a) Covering step.: Fix $\varepsilon > 0$ and let $\{g_1, \dots, g_N\}$ be an ε-net of $\mathcal{G}_\sigma$ under a chosen norm $\|\cdot\|$, with

$$N = N(\varepsilon, \mathcal{G}_\sigma, \|\cdot\|). \tag{17}$$

For any $g \in \mathcal{G}_\sigma$, choose g_j such that $\|g - g_j\| \le \varepsilon$. Then

$$\begin{aligned} & \left| \frac{1}{n} \sum_{k=T+1}^{T+n} g(X_k) - \mathbb{E} g(X_T) \right| \\ & \le \underbrace{\left| \frac{1}{n} \sum g_j(X_k) - \mathbb{E} g_j(X_T) \right|}_{\text{finite-class term}} + \underbrace{\left| \frac{1}{n} \sum (g - g_j)(X_k) \right|}_{\text{empirical approximation}} \\ & \quad + \underbrace{|\mathbb{E}(g - g_j)(X_T)|}_{\text{population approximation}}. \end{aligned} \tag{18}$$

b) Truncation and boundedness.: The class $\mathcal{G}_\sigma$ is not uniformly bounded on $\mathbb{R}^d$, so we localize to a high-probability event

$$E_R := \left\{ \max_{T+1 \le k \le T+n} \|X_k\| \le R \right\}, \tag{19}$$

with $R = R_\delta$ chosen using sub-Gaussian concentration of strongly log-concave laws (and similarly for π_h) [11].

On E_R, after centering test functions by constants, every $g \in \mathcal{G}_\sigma$ satisfies

$$|g(X_k)| \lesssim R + \sigma\sqrt{d}, \tag{20}$$

so the net elements are uniformly bounded by an envelope $b = b(R, \sigma, d)$.

c) Finite-class concentration with dependence.: Let γ_{ps} denote the pseudo spectral gap of the ULA chain (or the stationary chain after burn-in). Standard Markov-chain concentration inequalities imply that for each fixed bounded g_j,

$$\Pr\left(\left| \frac{1}{n} \sum g_j(X_k) - \pi_h(g_j) \right| > t \right) \lesssim \exp\left(-c\, n \gamma_{\mathrm{ps}} \frac{t^2}{b^2} \right), \tag{21}$$

for a universal constant $c > 0$ [10].

A union bound over the N net elements yields, with probability at least $1 - \delta$,

$$\begin{aligned} & \max_{j \le N} \left| \frac{1}{n} \sum g_j(X_k) - \pi_h(g_j) \right| \\ & \qquad \lesssim b \sqrt{\frac{\log N(\varepsilon, \mathcal{G}_\sigma, \|\cdot\|) + \log(1/\delta)}{n \gamma_{\mathrm{ps}}}}. \end{aligned} \tag{22}$$

The difference between $\pi_h(g_j)$ and $\mathbb{E} g_j(X_T)$ is controlled by the burn-in term from (11). After adding the approximation terms in (18) and optimizing over ε, this yields a baseline sampling bound in terms of the covering complexity of $\mathcal{G}_\sigma$.

E. Effective spectral dimension and covering complexity

The previous argument shows that the sampling term is governed by

$$\log N(\varepsilon, \mathcal{G}_\sigma, \|\cdot\|).$$

A crude worst-case estimate ignores the geometry of the target and leads to

$$\log N(\varepsilon, \mathcal{G}_\sigma, \cdot) \lesssim d \log(1/\varepsilon), \qquad W_1^{(\sigma)} \lesssim \sqrt{\frac{d}{n_{\mathrm{eff}}}}, \tag{23}$$

where $n_{\mathrm{eff}} \asymp n\gamma_{\mathrm{ps}}$.

Our goal is to replace the ambient dimension d by a spectral complexity term that reflects anisotropy of π_h. Let

$$\Sigma_h := \mathrm{Cov}_{\pi_h}(X), \tag{24}$$

with eigenvalues $\lambda_1(\Sigma_h), \dots, \lambda_d(\Sigma_h)$. Gaussian smoothing introduces a resolution scale σ, suppressing frequencies above scale $1/\sigma$ through

$$\widehat{g}(\omega) = \widehat{f}(\omega) e^{-\sigma^2 \|\omega\|^2/2}. \tag{25}$$

This suggests that directions with variance much smaller than σ^2 contribute weakly to the complexity of $\mathcal{G}_\sigma$ in $L_2(\pi_h)$.

Motivated by the Gaussian case (made precise below), we define the *effective spectral dimension*

$$d_{\log}(\sigma;\pi_h) := \log\det\Big(I + \tfrac{1}{\sigma^2}\Sigma_h\Big) = \sum_{i=1}^{d} \log\Big(1 + \tfrac{\lambda_i(\Sigma_h)}{\sigma^2}\Big). \tag{26}$$

We then posit the spectral covering bound

$$\log N(\varepsilon, \mathcal{G}_\sigma, L_2(\pi_h)) \lesssim d_{\log}(\sigma;\pi_h)\ \log(1/\varepsilon), \tag{27}$$

which is exact up to constants in the Gaussian setting and is the key structural refinement in our analysis.

A related quantity is the soft effective rank

$$d_{\mathrm{eff}}(\sigma;\pi_h) := \sum_{i=1}^{d} \frac{\lambda_i(\Sigma_h)}{\lambda_i(\Sigma_h) + \sigma^2}, \tag{28}$$

which provides an interpretable proxy for the number of directions visible above scale σ.

F. Gaussian and quadratic targets

a) Gaussian targets.: If $\pi = \mathcal{N}(0,\Sigma)$, the spectral covering bound (27) can be justified by an ellipsoidal frequency-counting argument: smoothing restricts the relevant Fourier modes to an ellipsoid determined by $\Sigma + \sigma^2 I$, and the logarithm of the corresponding volume yields

$$\log\det\Big(I + \tfrac{1}{\sigma^2}\Sigma\Big)$$

as the governing complexity term (cf. the geometric intuition behind Lemma 8). Consequently,

$$W_1^{(\sigma)} \lesssim \sqrt{\frac{d_{\log}(\sigma;\pi)}{n_{\mathrm{eff}}}}. \tag{29}$$

b) Quadratic approximation for general targets.: For general strongly log-concave targets, a natural strategy is to compare π locally with the Gaussian induced by the Hessian at the mode. Let x be the minimizer of V, and write

$$V(x) = V(x^*) + \tfrac{1}{2}(x - x^*)^\top H(x^*)(x - x^*) + R(x), \qquad H(x^*) = \nabla^2 V(x^*). \tag{30}$$

If $\nabla^2 V$ is Lipschitz with constant M, then the Taylor remainder obeys the cubic bound

$$|R(x)| \le \frac{M}{6}\|x - x^{\|^3}. \tag{31}$$

The quadratic part defines a Gaussian proxy with covariance $H(x^{)^{-1}}$. However, under strong log-concavity, typical fluctuations satisfy $\|X - x^{\|} \asymp \sqrt{d/\mu}$, so the remainder on typical samples scales as

$$|R(X)| \lesssim M\Big(\frac{d}{\mu}\Big)^{3/2}. \tag{32}$$

Therefore, transferring Gaussian entropy bounds to general targets requires a near-quadratic regime (e.g., $M(d/\mu)^{3/2} \ll 1$), or additional structure ensuring that the Laplace approximation remains accurate on the region carrying most of the mass.

G. Regimes of improvement

The proposed spectral complexity interpolates between isotropic and anisotropic regimes.

a) Isotropic regime.: If $\Sigma_h \approx \lambda I_d$, then

$$d_{\log}(\sigma;\pi_h) \approx d\ \log\Big(1 + \frac{\lambda}{\sigma^2}\Big), \tag{33}$$

so the resulting rate matches the standard dimension-driven bound up to logarithmic factors.

b) Anisotropic / low-effective-dimension regime.: If many eigenvalues satisfy $\lambda_i(\Sigma_h) \ll \sigma^2$, then their contributions to $d_{\log}(\sigma;\pi_h)$ are negligible and

$$d_{\log}(\sigma;\pi_h) \ll d. \tag{34}$$

In this case, the sampling term improves to

$$W_1^{(\sigma)} \lesssim \sqrt{\frac{d_{\log}(\sigma;\pi_h)}{n_{\mathrm{eff}}}}, \tag{35}$$

reflecting that only a subset of directions remains distinguishable at resolution σ.

IV. Conclusion and Future Work

In this work, we established a non-asymptotic bound for the smoothed Wasserstein distance that improves upon the classical sampling baseline in regimes where the effective dimension of the smoothed function class is significantly smaller than the ambient dimension. In particular, our analysis shows that by exploiting the covering structure of the smoothed class $\mathcal{G}_\sigma$, the statistical error can scale with a refined complexity measure rather than the worst-case dimensional dependence. This provides a more precise characterization of the sampling error in scenarios where smoothing reduces functional variability, leading to tighter guarantees than standard dimension-dependent bounds.

Several important directions remain for future investigation. First, it is of interest to understand how this intuition extends to broader classes of potential functions beyond the Gaussian and quadratic classes considered here, such as weakly smooth, anisotropic, or compositional function classes. A key technical challenge will be to identify and compute the appropriate intrinsic dimension or complexity measure governing these richer families. Second, it would be valuable to develop practical methods for estimating or bounding this effective dimension in concrete applications, which would enable sharper, data-dependent guarantees.

Finally, while our results focus on the Unadjusted Langevin Algorithm, it is important to investigate how similar complexity-based improvements can be obtained for other sampling methods, such as the Metropolis-Adjusted Langevin Algorithm (MALA) and Hamiltonian Monte Carlo (HMC). These methods exhibit different bias–variance and discretization trade-offs, and understanding how the function-class complexity interacts with their dynamics may lead to further improvements in sampling guarantees.

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Appendix A
Proofs for Background Claims

Theorem Lemma

A. Brownian roughness: why the SDE cannot be simulated exactly

Lemma 1 (Variance blow-up of Brownian finite differences)**.** *Let $(B_t)_{t\geq 0}$ be standard 1D Brownian motion. Then for any $t \geq 0$ and $h > 0$,*

$$\frac{B_{t+h} - B_t}{h} \sim \mathcal{N}\left(0, \frac{1}{h}\right).$$

In particular, the variance of the slope estimator diverges as $h \to 0$.

Proof. Brownian increments satisfy $B_{t+h} - B_t \sim \mathcal{N}(0, h)$. Dividing by h gives a centered Gaussian with variance $h/h^2 = 1/h$. □

Theorem 1 (Brownian motion is almost surely nowhere differentiable)**.** *With probability one, the sample path $t \mapsto B_t$ is nowhere differentiable.*

Proof. A standard proof uses quadratic variation. For any partition $0 = t_0 < t_1 < \cdots < t_n = t$ with mesh $\max_i(t_{i+1} - t_i) \to 0$, one has

$$\sum_{i=0}^{n-1} (B_{t_{i+1}} - B_{t_i})^2 \to t \quad \text{almost surely.}$$

If B_t were differentiable at some $t^\star$, then locally $B_{t^\star+h} - B_{t^\star} = B'(t^\star)h + o(h)$, implying the squared increment is $B'(t^\star)^2 h^2 + o(h^2)$, and the quadratic variation around $t^\star$ would scale like $\sum h^2$, which vanishes as mesh $\to 0$, contradicting nonzero quadratic variation. Hence no point of differentiability exists almost surely. □

B. Euler–Maruyama discretization: deriving ULA

Lemma 2 (Euler–Maruyama discretization of Langevin gives ULA)**.** *Let X_t solve $dX_t = -\nabla V(X_t)\,dt + \sqrt{2}\,dB_t$ and let $t_k = kh$. Formally integrating over $[t_k, t_{k+1}]$ yields*

$$X_{t_{k+1}} - X_{t_k} = -\int_{t_k}^{t_{k+1}} \nabla V(X_t)\,dt + \sqrt{2}\,(B_{t_{k+1}} - B_{t_k}).$$

Approximating the drift integral by $h\nabla V(X_{t_k})$ gives the ULA update

$$X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0, I_d).$$

Proof. The integral identity is the Itô form of the SDE. Brownian increments satisfy $B_{t_{k+1}} - B_{t_k} \sim \mathcal{N}(0, hI_d)$ and can be written as $\sqrt{h}\,\xi_k$ with $\xi_k \sim \mathcal{N}(0, I_d)$. Replacing the drift integral by $h\nabla V(X_{t_k})$ gives the stated recursion. □

C. Stationary distribution mismatch: the quadratic example

Lemma 3 (Stationary law of ULA for a quadratic potential)**.** *Let $V(x) = \frac{1}{2}\|x\|^2$ so $\pi = \mathcal{N}(0, I_d)$. ULA becomes*

$$X_{k+1} = (1-h)X_k + \sqrt{2h}\,\xi_k, \qquad \xi_k \sim \mathcal{N}(0, I_d),$$

with $h \in (0, 2)$. Then the chain admits a unique stationary distribution

$$\pi_h = \mathcal{N}(0,\ aI_d)\,, \qquad a = \frac{2}{2-h}.$$

In particular, $\pi_h \neq \pi$ for any $h > 0$.

Proof. This is a linear Gaussian autoregression. If $X_k \sim \mathcal{N}(0, \Sigma)$ then

$$X_{k+1} \sim \mathcal{N}\big(0,\ (1-h)^2\Sigma + 2hI_d\big)\,.$$

Stationarity requires $\Sigma = (1-h)^2\Sigma + 2hI_d$, i.e.

$$\big(1 - (1-h)^2\big)\Sigma = 2hI_d.$$

Since $1 - (1-h)^2 = 2h - h^2 = h(2-h)$, we obtain

$$\Sigma = \frac{2h}{h(2-h)} I_d = \frac{2}{2-h} I_d.$$

Existence and uniqueness follow because $|(1-h)| < 1$ when $h \in (0, 2)$. □

D. W_2 contraction for ULA under strong convexity

Lemma 4 (One-step contraction under synchronous coupling)**.** *Assume V is m-strongly convex and L-smooth:*

$$mI_d \preceq \nabla^2 V(x) \preceq LI_d.$$

Let (X_k) and (Y_k) follow ULA with the same Gaussian noise ξ_k:

$$X_{k+1} = X_k - h\nabla V(X_k) + \sqrt{2h}\,\xi_k, \quad Y_{k+1} = Y_k - h\nabla V(Y_k) + \sqrt{2h}\,\xi_k.$$

Let $\Delta_k := X_k - Y_k$. Then

$$\|\Delta_{k+1}\|^2 \leq \big(1 - 2mh + h^2L^2\big)\,\|\Delta_k\|^2.$$

In particular, if $h \leq m/L^2$, then

$$\|\Delta_{k+1}\| \leq (1 - mh)\,\|\Delta_k\|.$$

Proof. Subtract the recursions to cancel noise:

$$\Delta_{k+1} = \Delta_k - h\big(\nabla V(X_k) - \nabla V(Y_k)\big).$$

Expand:

$$\begin{aligned}\|\Delta_{k+1}\|^2 = \|\Delta_k\|^2 - 2h\langle \Delta_k, \nabla V(X_k) - \nabla V(Y_k)\rangle \\ + h^2\|\nabla V(X_k) - \nabla V(Y_k)\|^2.\end{aligned}$$

Strong convexity implies monotonicity:

$$\langle \Delta_k, \nabla V(X_k) - \nabla V(Y_k)\rangle \geq m\|\Delta_k\|^2.$$

Smoothness implies Lipschitz gradient:

$$\|\nabla V(X_k) - \nabla V(Y_k)\| \leq L\|\Delta_k\|.$$

Substitute to obtain

$$\|\Delta_{k+1}\|^2 \le \big(1 - 2mh + h^2L^2\big)\|\Delta_k\|^2.$$

If $h \le m/L^2$, then $1 - 2mh + h^2L^2 \le (1 - mh)^2$, giving the norm contraction. □

[W_2 contraction of the ULA kernel] Let P_h be the Markov kernel of ULA. Under the step-size condition above, for any laws μ, ν on ^{d},

$$W_2(\mu P_h, \nu P_h) \le (1 - mh)\, W_2(\mu, \nu).$$

Proof. Couple $X_0 \sim \mu$ and $Y_0 \sim \nu$ optimally for W_2, then run synchronous coupling. Take expectations and use the previous lemma:

$$\|X_1 - Y_1\| \le (1 - mh)\, \|X_0 - Y_0\|.$$

Minimize over couplings to get the W_2 statement. □

E. Local-to-global control of $W_2(\pi_h, \pi)$

Lemma 5 (Local coupling estimate for one step)**.** *Assume V is L-smooth. Let $Q_h(x, \cdot)$ denote the law of the exact diffusion at time h starting from x, and let $P_h(x, \cdot)$ denote the ULA one-step kernel starting from x. Then, coupling both using the same Brownian increment,*

$$W_2\big(Q_h(x, \cdot), P_h(x, \cdot)\big) \le C(L)\, h^{3/2},$$

where $C(L)$ is a constant depending on L (and mild moment bounds).

Proof. Let X_t solve the SDE with $X_0 = x$ and define the ULA proposal

$$\widetilde{X}_h = x - h\nabla V(x) + \sqrt{2}\,(B_h - B_0).$$

Then

$$X_h - \widetilde{X}_h = -\int_0^h \big(\nabla V(X_t) - \nabla V(x)\big)\, dt.$$

By L-smoothness,

$$\|X_h - \widetilde{X}_h\| \le \int_0^h L\|X_t - x\|\, dt.$$

Using Cauchy–Schwarz,

$$\|X_h - \widetilde{X}_h\|^2 \le L^2 h \int_0^h \|X_t - x\|^2\, dt.$$

For the SDE, $\|X_t - x\|^2 = O(t)$ for small t (it is dominated by the Brownian term), so $\int_0^h \|X_t - x\|^2 dt = O(h^2)$. Hence

$$\|X_h - \widetilde{X}_h\|^2 = O(h^3), \quad \|X_h - \widetilde{X}_h\| = O(h^{3/2}).$$

Since this is an explicit coupling, it upper bounds W_2. □

Lemma 6 (Global stationary bias bound from local error + contraction)**.** *Assume (i) W_2-contraction of P_h with factor $(1-mh)$ and (ii) a local bound $\sup_x W_2(Q_h(x, \cdot), P_h(x, \cdot)) \le Ch^{3/2}$. Let π be invariant for Q_h and let π_h be invariant for P_h. Then*

$$W_2(\pi, \pi_h) \le \frac{C}{m}\, h^{1/2}.$$

Proof. Use invariance $\pi = \pi Q_h$ and $\pi_h = \pi_h P_h$ and write

$$\begin{aligned} W_2(\pi, \pi_h) &= W_2(\pi Q_h, \pi_h P_h) \\ &\le W_2(\pi Q_h, \pi P_h) + W_2(\pi P_h, \pi_h P_h). \end{aligned}$$

The first term is at most $Ch^{3/2}$ by integrating the pointwise local bound over $x \sim \pi$. The second term is at most $(1 - mh)W_2(\pi, \pi_h)$ by W_2 contraction. Rearrange:

$$mh\, W_2(\pi, \pi_h) \le Ch^{3/2},$$

which yields the claim. □

F. Explicit W_2 for the quadratic Gaussian example

Lemma 7 (Wasserstein–2 distance between centered isotropic Gaussians)**.** *Let $\mathcal{N}(0, I_d)$ and $\mathcal{N}(0, aI_d)$ with $a > 0$. Then*

$$W_2\big(\mathcal{N}(0, I_d), \mathcal{N}(0, aI_d)\big) = \sqrt{d}\,|\sqrt{a} - 1|.$$

Proof. For centered Gaussians, the optimal transport map is linear and the closed form is

$$W_2^2 = (\Sigma_1) + (\Sigma_2) - 2\Big((\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2})^{1/2}\Big).$$

With $\Sigma_1 = I_d$ and $\Sigma_2 = aI_d$, we have $\Sigma_2^{1/2}\Sigma_1\Sigma_2^{1/2} = aI_d$ and its square root is $\sqrt{a}I_d$. Thus

$$W_2^2 = d + ad - 2\sqrt{a}\, d = d(\sqrt{a} - 1)^2,$$

giving the result. □

[Small-h expansion for ULA stationary bias in the quadratic case] For $V(x) = \frac{1}{2}\|x\|^2$, $\pi = \mathcal{N}(0, I_d)$ and $\pi_h = \mathcal{N}\Big(0, \frac{2}{2-h} I_d\Big)$. Then

$$W_2(\pi_h, \pi) = \sqrt{d}\left(\sqrt{\frac{2}{2-h}} - 1\right) = \frac{\sqrt{d}}{4}\, h + O(h^2) \quad (h \to 0).$$

Proof. Apply the previous lemma with $a = 2/(2 - h)$ and Taylor expand $(1 - h/2)^{-1/2} = 1 + \frac{h}{4} + O(h^2)$. □

G. Rejection sampling: exactness

Theorem 2 (Correctness of rejection sampling)**.** *Let $\tilde{\pi}$ and $\tilde{\mu}$ be nonnegative functions with $\tilde{\pi}(x) \le \tilde{\mu}(x)$ for all x. Let μ be the distribution with density $\tilde{\mu}/Z_\mu$. Sample $X \sim \mu$ and $U \sim \mathrm{Unif}[0, 1]$ independently and accept X if*

$$U \le \frac{\tilde{\pi}(X)}{\tilde{\mu}(X)}.$$

Then the accepted sample has distribution π with density $\tilde{\pi}/Z_\pi$.

Proof. Let $(X_i, U_i)_{i \ge 1}$ be i.i.d. copies of (X, U) and let T be the first accepted index. For any measurable set A,

$$\begin{aligned} (X_T \in A) &= \sum_{n \ge 0} \Big(X_{n+1} \in A,\ U_{n+1} \le \tfrac{\tilde{\pi}(X_{n+1})}{\tilde{\mu}(X_{n+1})},\ \text{first } n \text{ rejected}\Big) \\ &= \sum_{n \ge 0} \Big(X_{n+1} \in A,\ U_{n+1} \le \tfrac{\tilde{\pi}(X_{n+1})}{\tilde{\mu}(X_{n+1})}\Big)(\text{reject})^n, \end{aligned}$$

by independence across trials. Now

$$(\text{accept}) = \left[\frac{\tilde{\pi}(X)}{\tilde{\mu}(X)}\right] = \int \frac{\tilde{\pi}(x)}{\tilde{\mu}(x)} \cdot \frac{\tilde{\mu}(x)}{Z_\mu}\, dx = \frac{Z_\pi}{Z_\mu}.$$

Similarly,

$$\Big(X \in A,\ \text{accept}\Big) = \int_A \frac{\tilde{\pi}(x)}{Z_\mu}\, dx = \frac{Z_\pi}{Z_\mu}\, \pi(A).$$

Therefore

$$(X_T \in A) = \sum_{n\geq 0} \frac{Z_\pi}{Z_\mu}\, \pi(A) \left(1 - \frac{Z_\pi}{Z_\mu}\right)^n = \pi(A).$$

□

H. Metropolis–Hastings (incl. MALA) detailed balance

Lemma 8 (Detailed balance for Metropolis–Hastings)**.** *Let π be a target density and $q(x \to y)$ a proposal density. Define*

$$A(x,y) = \min\left\{1,\ \frac{\pi(y)\, q(y \to x)}{\pi(x)\, q(x \to y)}\right\}.$$

Then for $x \neq y$,

$$\pi(x)\, q(x \to y)\, A(x,y) = \pi(y)\, q(y \to x)\, A(y,x).$$

Proof. Let $r(x,y) = \frac{\pi(y)\, q(y\to x)}{\pi(x)\, q(x\to y)}$. Then

$$\pi(x)q(x \to y)A(x,y) = \pi(x)q(x \to y)\min\{1, r(x,y)\} = \min\{\pi(x)q(x \to y),\ \pi(y)q(y \to x)\},$$

which is symmetric in (x,y) and equals $\pi(y)q(y \to x)A(y,x)$. □

I. Markov concentration (bounded case) and the unboundedness fix

Theorem 3 (Bounded-function Markov concentration (Paulin))**.** *Let (X_k) be a Markov chain with stationary distribution π_h and pseudo-spectral gap $\gamma_{\rm ps} > 0$ (as in [10]). If $\|g\|_\infty \leq b$, then there is a universal $c > 0$ such that for all $t > 0$,*

$$\left(\left|\frac{1}{n}\sum_{k=1}^n g(X_k) - \pi_h(g)\right| > t\right) \leq 2\exp\left(-c\, n\, \gamma_{\rm ps} \frac{t^2}{b^2}\right).$$

Lemma 9 (Sub-Gaussian tails for Lipschitz observables under LSI)**.** *Assume π_h satisfies a log-Sobolev inequality with constant $C_{\rm LSI}$. Then for any 1-Lipschitz f and all $u \geq 0$,*

$$\pi_h(|f -_{\pi_h} f| \geq u) \leq 2\exp\left(-\frac{u^2}{2C_{\rm LSI}}\right).$$

Proof. This is the standard Herbst argument; see [1, Chapter 2]. □

Lemma 10 (Truncation reduction for unbounded Lipschitz functions)**.** *Let (X_k) be stationary with invariant distribution π_h and pseudo-spectral gap $\gamma_{\rm ps} > 0$. Let g be 1-Lipschitz and assume π_h satisfies LSI($C_{\rm LSI}$). For $b > 0$, define $g^{(b)}(x) = \max\{-b, \min\{g(x), b\}\}$. Then for any $t > 0$,*

$$\left(\left|\frac{1}{n}\sum_{k=1}^n g(X_k) - \pi_h(g)\right| > t\right) \leq 2\exp\left(-c\, n\, \gamma_{\rm ps} \frac{t^2}{b^2}\right) + 4n\exp\left(-\frac{(b-\Delta)^2}{2C_{\rm LSI}}\right),$$

where c is from Theorem 3 and $\Delta := |\pi_h(g)|$.

Proof. Write

$$\frac{1}{n}\sum_{k=1}^n g(X_k) - \pi_h(g) = \left(\frac{1}{n}\sum_{k=1}^n g^{(b)}(X_k) - \pi_h(g^{(b)})\right) + R,$$

where

$$R := \left(\frac{1}{n}\sum_{k=1}^n (g - g^{(b)})(X_k) - \pi_h(g - g^{(b)})\right).$$

By the union bound,

$$(|A + R| > t) \leq (|A| > t/2) + (|R| > t/2).$$

Since $|g^{(b)}| \leq b$, Theorem 3 gives

$$(|A| > t/2) \leq 2\exp\left(-c\, n\, \gamma_{\rm ps} \frac{t^2}{b^2}\right),$$

after absorbing constants into c.

For the remainder, note that $(g - g^{(b)})(x) = 0$ whenever $|g(x)| \leq b$. Hence,

$$|R| > 0 \ \Rightarrow\ \exists k \leq n:\ |g(X_k)| > b.$$

Thus, by a union bound and stationarity,

$$\begin{aligned}(|R| > t/2) &\leq (\exists k \leq n:\ |g(X_k)| > b)\\ &\leq n\, \pi_h(|g| > b).\end{aligned}$$

If $|g| > b$ then $|g - \pi_h(g)| > b - \Delta$, so Lemma 9 yields

$$\pi_h(|g| > b) \leq 2\exp\left(-\frac{(b-\Delta)^2}{2C_{\rm LSI}}\right).$$

Combining bounds and adding slack gives the stated second term. □

Remark 1. *Choosing $b \asymp \sqrt{C_{\rm LSI}\log(n/\delta)}$ makes the tail term $4n\exp(-(b-\Delta)^2/(2C_{\rm LSI}))$ of order δ.*